SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Salt-and-Pepper Noise
A scanned document contains salt-and-pepper noise affecting readability.
(a) Explain the scenario and characteristics of this noise.
(b) Identify key issues impacting image quality.
(c) Apply an appropriate filtering technique.
(d) Justify why this method is more suitable than alternatives.
(e) Present your response clearly and logically.

(a) Scenario & Characteristics:
Salt-and-pepper noise appears as random black and white pixels in a scanned document, reducing readability.

(b) Key Issues:

* Poor readability
* Loss of text clarity
* Random pixel corruption
* Reduced image quality

(c) Suitable Filtering Technique:
Apply a Median Filter.

(d) Justification:
The median filter removes salt-and-pepper noise effectively while preserving image edges. It performs better than the mean filter because it does not blur the image.

(e) Conclusion:

Median filtering restores document quality and improves readability.

2. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario and explain the importance of boundary detection.

(b) Identify key challenges in detecting boundaries.

(c) Apply a suitable edge detection method.

(d) Justify your selection with logical reasoning.

(e) Present your explanation clearly and systematically.

(a) Scenario & Importance:

In medical images, accurate boundary detection helps distinguish different tissues for diagnosis.

(b) Key Challenges:

* Noise
* Weak edges
* Low contrast
* Complex tissue structures

(c) Suitable Method:

Use the Canny Edge Detection algorithm.

(d) Justification:

Canny detects edges accurately, reduces noise, and provides clear boundaries compared to Sobel or Prewitt.

(e) Conclusion:

Canny edge detection provides reliable and accurate tissue boundary detection.

3. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements.
(b) Identify key features enabling separation.
(c) Apply a suitable segmentation technique.
(d) Justify your decision logically.
(e) Present your response clearly.

(a) Scenario:
Objects are easily separated from the background because they have different intensity values.

(b) Key Features:

* High intensity difference
* Uniform background
* Good contrast

(c) Suitable Technique:
Threshold Segmentation (Global Thresholding)

(d) Justification:
Thresholding is simple, fast, and effective when object and background intensities are clearly different.

(e) Conclusion:
Threshold segmentation efficiently separates objects from the background.

4. Low Brightness Image

An image captured in low light appears very dark with poor visibility.
(a) Interpret the scenario and explain the cause of low brightness.
(b) Identify key issues affecting image visibility.
(c) Apply an appropriate enhancement technique.
(d) Justify your choice with reasoning.
(e) Present your explanation clearly.

(a) Scenario:

The image is dark due to insufficient lighting during image capture.

(b) Key Issues:

* Poor visibility

* Low contrast

* Hidden details

(c) Enhancement Technique:

Apply Histogram Equalization.

(d) Justification:

Histogram Equalization improves brightness and contrast better than simple brightness adjustment.

(e) Conclusion:

The enhanced image becomes brighter and easier to analyze.

5. Gaussian Noise

An image contains Gaussian noise affecting smooth regions.

(a) Interpret the scenario and describe Gaussian noise.

(b) Identify key issues affecting quality.

(c) Apply an appropriate smoothing filter.

(d) Justify your method over alternatives.

(e) Present your explanation clearly.

(a) Scenario:

Gaussian noise appears as random intensity variations caused by sensors or electronic interference.

(b) Key Issues:

* Grainy appearance

* Reduced image quality

* Difficulty in feature extraction

(c) Suitable Filter:

Gaussian Filter

(d) Justification:

The Gaussian filter smooths noise while preserving overall image structure better than average filtering.

(e) Conclusion:

Gaussian filtering improves image quality and prepares the image for further processing.

6. Loss of Fine Details

An image loses fine details after processing.

(a) Interpret the scenario and explain detail loss.

(b) Identify key issues related to filtering.

(c) Apply a suitable enhancement technique.

(d) Justify your method with reasoning.

(e) Present your explanation clearly.

(a) Scenario:

Fine image details disappear after excessive smoothing.

(b) Key Issues:

* Blurred edges
* Loss of texture
* Reduced sharpness

(c) Enhancement Technique:

Apply Unsharp Masking (Image Sharpening).

(d) Justification:

Unsharp masking restores fine details and improves edge sharpness without excessive distortion.

(e) Conclusion:

The image becomes sharper and important details are preserved.

7. Edge Enhancement Requirement

An application requires highlighting edges for feature extraction.

(a) Interpret the scenario and explain the need for edge enhancement.
(b) Identify key issues in detecting edges.
(c) Apply an appropriate method.
(d) Justify your choice logically.
(e) Present your explanation clearly.

(a) Scenario:
Edges need to be highlighted for accurate feature extraction in computer vision applications.

(b) Key Issues:

* Weak edges
* Noise interference
* Low contrast

(c) Suitable Method:
Use the Sobel Edge Detection method.

(d) Justification:
Sobel enhances edges while maintaining computational efficiency and is suitable for feature extraction.

(e) Conclusion:
Sobel edge enhancement makes object boundaries more visible for further analysis.

Code of Conduct:

I VIGNESH/(192525013)*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:VIGNESH

RUBRICS FOR CALCULATION:

Criteria	Weight ag e (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Unders ta nding of Scenari o	20%	Demonstrat e s clear and complete understandi n g of the scenario and context	Shows good understandi n g with minor gaps	Partial understan di ng; misses key aspects	Misinterpr et s or fails to understan d the scenario
Identific at ion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicati o n of Concept s	25%	Applies appropriat e concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decisi on Making	20%	Makes well- reasone d, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity o f Respo ns e	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewh at unclear or poorly structure d	Disorganiz e d and difficult to understan d